Electronics Networks and Instruments & Measurements

Wireless PSI (Class-2 / Class-3) Exam-Oriented ટૂંકી નોંધ (Gujarati)

📘 ભાગ A: NETWORK THEOREMS

Theorem	મુખ્ય વાત	ક્યાં વપરાય
Superposition	એક-એક Source Active રાખો, બાકીના Voltage Source=Short, Current Source=Open. Results Algebraically ઉમેરો	Multiple Sources, **Linear** Circuit. **Power માટે ક્યારેય નહીં** (Non-linear)
Thevenin	Network = Vth (Series Rth)	Load બદલાય ત્યારે Repeated Calculation માટે Best
Norton	Network = IN (Parallel RN); RN = Rth	Thevenin નું Current-source સ્વરૂપ
Max Power Transfer	RL = Rth (DC); ZL = Zth* (AC)	Efficiency ફક્ત 50%; Signal Circuits માં વપરાય, Power System માં નહીં
Millman	અનેક Parallel Voltage Sources → 1 Equivalent Source	Parallel Source Combine કરવા
Reciprocity	Source-Meter Position Swap → Ratio Same રહે	ફક્ત Single Source, Linear, Bilateral Network
Compensation	ΔR ફેરફાર = નવો Voltage Source (I×ΔR) ઉમેરાય, બાકી Sources Zero	Resistance Variation Analysis
Tellegen	Σ(Vk×Ik) = 0 (બધા Branch Power નો Sum = 0)	**સૌથી General** - Linear/Non-linear, Active/Passive, Time-variant બધા માટે લાગુ

🧠 યાદ રાખવાની Trick

- **"SToNo MiR CT"** — Superposition, Thevenin, Norton, Millman, Reciprocity, Compensation, Tellegen
- Tellegen જ એક એવો theorem છે જેને Linearity ની જરૂર નથી.
- Formula: $P_{max} = V_{th}^2/4R_{th}$

⚠️ Common Exam Trap

- Superposition થી Power ક્યારેય ન ગણવી.

- Reciprocity ફક્ત Single Source Network માટે.

📘 ભાગ B: RESONANCE (રેઝોનન્સ)

Series Resonance (Acceptor Circuit / Voltage Resonance)

Parameter	Formula/Behavior
Condition	XL = XC
f_0	$1/(2\pi\sqrt{LC})$
Z	Minimum = R
I	Maximum = V/R
Q-Factor	$(1/R)\sqrt{(L/C)} = \omega_0 L/R$
VL = VC	Q × Vsource
Bandwidth	f_0/Q

Parallel Resonance (Rejector Circuit / Current Resonance)

- Z = Maximum, I (Line) = Minimum
- Dynamic Resistance Rd = L/(CR)
- Circulating Current L-C Branch માં Large હોય છે (Q ગણો)

Series vs Parallel — Quick Compare

Point	Series	Parallel
Impedance	Min	Max
Current	Max	Min
નામ	Acceptor	Rejector
Voltage Magnification	હા	ના
Current Magnification	ના	હા
ઉપયોગ	Radio Tuning, Filter	Tank Circuit, Oscillator

🧠 **Trick: "SAM-PRC" → Series=Acceptor=Max Current, Parallel=Rejector=Current(circulating) magnified**

📘 ભાગ C: COUPLED CIRCUITS

- Mutual Inductance: $M = k\sqrt{(L1L2)}$
- Coefficient of Coupling $k = M/\sqrt{(L1L2)}$, range $0 \leqslant k \leqslant 1$ (k=1 → Ideal Transformer)
- $k<0.5$ = Loose, $k>0.5$ = Tight
- **Series Aiding:** $Leq = L1+L2+2M$
- **Series Opposing:** $Leq = L1+L2-2M$
- **M (Experimental)** = (Laiding − Lopposing)/4
- **Dot Convention:** Same Dot Entry → +M (Additive); Opposite → −M (Subtractive)
- Tuned Coupled Circuit: Loose k → 1 peak; Critical k → Flat top; Tight k → Double hump

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📘 ભાગ D: ATTENUATOR, EQUALIZER, FILTER

Attenuator (dB માપન: $20 \log_{10}(Vout/Vin)$)

Type	લક્ષણ
T-type	3 Resistor, T-shape, Symmetrical
π-type	3 Resistor, π-shape
L-type	Asymmetrical, simplest
Bridged-T	Variable Attenuation
Ladder	Cascaded L-sections, step variable (Test Instruments)

Equalizer

- Frequency-dependent Loss compensate કરી Flat Response આપે
- Types: **Amplitude**, **Phase(Delay)**, **Lattice** (Constant-R), **Bridged-T**, Active/Passive
- Constant Resistance Condition: $Z1 \times Z2 = R0^2$

Filter vs Attenuator vs Equalizer

	કાર્ય
Attenuator	બધી Frequency ને Equally ઘટાડે

	કાર્ય
Equalizer	Loss Compensate કરી Flat બનાવે
Filter	Frequency-Selective — અમુક Pass, અમુક Block

Filter Types (Frequency-wise)

Type	Pass Band
LPF	0 - fc
HPF	fc - ∞
BPF	f1 - f2
BSF (Notch)	બધું, સિવાય f1-f2

Design-wise Filter

Type	લક્ષણ
Constant-K	સાદો, Slow Roll-off, $f_c = 1/(\pi\sqrt{LC})$
m-Derived	Sharp Cut-off, $f_\infty = f_c/\sqrt{(1-m^2)}$, સામાન્ય m=0.6
Composite	Constant-K + m-Derived નું Combination

Active Filter Response Types

Filter	લક્ષણ
Butterworth	Maximally Flat (No Ripple)
Chebyshev	Ripple + Sharp Roll-off
Bessel	Linear Phase (Pulse Signal Best)
Elliptic	બંને Band માં Ripple, સૌથી Sharp

🧠 **Trick: "LoBAM" - LPF, HPF, BPF, BSF (Low-High-Band Pass-Band Stop) ક્રમમાં યાદ રાખો**

📘 ભાગ E: TRANSMISSION LINE THEORY

Primary Constants (Per Unit Length)

R (Ω/km), L (H/km), C (F/km), G (S/km — Mho/km)

Secondary Constants

Constant	Formula
Characteristic Impedance Z0 (Lossless)	$\sqrt{(L/C)}$
Propagation Constant γ	$\alpha + j\beta = \sqrt{[(R+j\omega L)(G+j\omega C)]}$
Velocity	$1/\sqrt{(LC)}$ (lossless)

Special Line Types

Type	Condition
Lossless	R=0, G=0
Distortionless	R/L = G/C

Reflection

- $\Gamma = (ZL-Z0)/(ZL+Z0)$
- $\Gamma=0$ → Matched (ZL=Z0); $\Gamma=-1$ → Short; $\Gamma=+1$ → Open
- VSWR: $S = (1+|\Gamma|)/(1-|\Gamma|)$; Range $1 \leq S \leq \infty$; S=1 = Perfect Matching

Special Lines

Line	Zin
Open Circuit	$-jZ0 \cot(\beta l)$
Short Circuit	$jZ0 \tan(\beta l)$
Matched (ZL=Z0)	Z0
Quarter-wave ($\lambda/4$)	$Z0^2/ZL$ — **Impedance Inverter**
Half-wave ($\lambda/2$)	ZL — **Impedance Repeater**

🧠 Must Remember

1. Matched Line → $\Gamma=0$ → VSWR=1 → Max Power Transfer

2. Distortionless Condition: R/L = G/C (Common MCQ)
3. Quarter-wave → Invert, Half-wave → Repeat
4. Loss Types: Ohmic, Dielectric, Radiation, Corona, Skin Effect

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📘 ભાગ F: NETWORK TOPOLOGY (GRAPH THEORY)

Term	વ્યાખ્યા
Node	જ્યાં 2+ Elements મળે
Branch	2 Nodes વચ્ચેનું Connection
Tree	બધા Nodes Include, કોઈ Loop નહીં, બધા Connected
Twig	Tree ની Branch, Number = n−1
Link/Chord	Co-Tree ની Branch, Number = b−n+1

Key Relations (n=Nodes, b=Branches)

- Independent Loops = $b - n + 1$
- Twigs = $n - 1$
- Links = $b - n + 1$

Matrices

Matrix	ઉપયોગ	Size
Incidence Matrix (A)	Node-Branch સંબંધ (દરેક Column Sum=0)	(n−1)×b (Reduced)
Tie-Set Matrix (B)	KVL → [B][V]=0	Rows=Links
Cut-Set Matrix (Q)	KCL → [Q][I]=0	Rows=Twigs

Duality

Original	Dual
Series	Parallel
V	I
R	G

Original	Dual
L	C
KVL	KCL
Mesh	Node
Short Circuit	Open Circuit

🧠 Trick: "Tie=KVL(Loop), Cut=KCL(Node)" — T પહેલા આવે તો Loop, C પહેલા આવે તો Current(KCL)

📘 ભાગ G: INSTRUMENTS INTRO — Errors & Standards

Term	અર્થ
Accuracy	True Value ની નજીક
Precision	Repeatability (Consistent Results)
Resolution	સૌથી નાનો Detectable Change
Sensitivity	Output ફેરફાર / Input ફેરફાર

Errors

Type	કારણ
Gross Error	Human Mistake
Systematic Error	Instrumental/Environmental/Observational (Consistent Pattern)
Random Error	Unpredictable, Statistical Method થી ઘટાડાય

- % Error = (Absolute Error/True Value)×100

Standards Hierarchy

International → Primary → Secondary → Working

DC Bridges

Bridge	ઉપયોગ	Balance Condition
Wheatstone	Medium R (1Ω-1MΩ)	P×S = Q×R
Kelvin's Double	Low R (<1Ω)	P/Q = p/q (Double Ratio Arms)
Megger	High R / Insulation (MΩ)	Hand/Battery driven HV Generator

AC Bridges — સૌથી Important Table (Exam Favorite)

Bridge	Measure કરે	Standard Component
Maxwell's L-L	Medium Q Inductance	Standard Inductor
Maxwell's L-C	Medium Q Inductance	Standard Capacitor (1<Q<10)
Hay's Bridge	High Q Inductance (Q>10)	Standard Capacitor (Series R)
Anderson's Bridge	Inductance, any Q, High Accuracy	Standard Capacitor (Extra R)
Schering Bridge	Capacitance + Dielectric Loss (tan δ)	Standard Capacitor — **સૌથી વધુ વપરાતું**
Wien's Bridge	Frequency Measurement	R & C બંને
De Sauty's Bridge	Capacitance Comparison (Loss-free only)	Standard Capacitor

- General AC Balance: Z1×Z4 = Z2×Z3 (Magnitude + Phase બંને Satisfy)
- Schering: tan δ = ωR4C4
- Wien's: f = 1/(2π√(R2R4C2C4))

🧠 Trick: "MHASWD" - Maxwell, Hay's, Anderson, Schering, Wien, De Sauty (ક્રમમાં યાદ)

- Maxwell fails for High Q → Hay's solves it
- Anderson = Best Accuracy (Extra R, Frequency Independent)
- Schering = **Capacitance ka King**, Wien = **Frequency ka King**

📘 ભાગ I: BASIC PARAMETER MEASUREMENT

Parameter	Best Method
Low Resistance	Kelvin's Double Bridge
Medium Resistance	Wheatstone Bridge
High Resistance	Megger
Inductance (Low Q)	Maxwell's Bridge
Inductance (High Q)	Hay's Bridge
Capacitance	Schering / De Sauty's Bridge
Frequency	Wien's Bridge
Power	Dynamometer Wattmeter ($P=VI\cos\varphi$)
Energy	Induction Type Energy Meter
Q-Factor	Q-Meter ($Q = VC/V$)

- Ammeter-Voltmeter Method: Ammeter First = Low R માટે; Voltmeter First = High R માટે

📘 ભાગ J: OSCILLOSCOPE (CRO)

CRT ના મુખ્ય Parts (ક્રમ યાદ રાખો)

Electron Gun → Deflection Plates (Y then X) → Fluorescent Screen

Block	કાર્ય
Vertical Amplifier	Input signal ને Amplify (Y-plate)
Delay Line	Vertical Signal ને Delay કરે (Initial part cut ન થાય)
Time Base Generator	**Sawtooth Wave** બનાવે (Horizontal Sweep)
Trigger Circuit	Waveform ને Steady કરે (Sync)
Horizontal Amplifier	Time base ને Amplify (X-plate)

Trigger Modes

Mode	વર્ણન
Auto	Signal ના હોય તો પણ Sweep ચાલે
Normal	ફક્ત Trigger Condition Satisfy થાય ત્યારે
Single	એક જ વાર Capture

Measurements

- V = Divisions × Volts/Div
- T = Divisions × Time/Div; f = 1/T
- Phase (Lissajous): $\sin\varphi = Y1/Y2$
- Frequency Ratio (Lissajous) = Vertical Tangency/Horizontal Tangency

CRO Types

Type	ખાસિયત
Analog CRO	Basic
Dual Trace	Alternate mode=High freq, Chopped mode=Low freq
Storage CRO	લાંબો સમય Store
DSO	ADC વડે Digital Store, Freeze, Pre-trigger Capture
Sampling Scope	GHz Range, Sample-by-sample Reconstruct

🧠 Trick: "TB-Sawtooth, always" — Time base = Sawtooth (100% MCQ ask)

📘 ભાગ K: SIGNAL GENERATORS

Type	Frequency Range	Waveform
AF Generator	20Hz-20kHz	Sine
RF Generator	kHz-GHz	Sine (AM/FM Modulation possible)
Function Generator	Hz-MHz	Sine, Square, Triangle, Sawtooth (**Most Versatile**)

Type	Frequency Range	Waveform
Pulse Generator	—	Rectangular Pulses
Sweep Generator	Auto varying	Frequency Response Test
Arbitrary Waveform Gen (AWG)	Digital	Any Custom Shape

- Function Generator Working: Integrator (Triangular) → Schmitt Trigger (Square) → Shaping (Sine)
- Requirements: Frequency Stability, Frequency Accuracy, Amplitude Stability, Low Distortion, Wide Range

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📘 ભાગ L: FREQUENCY COUNTER

Block Order (યાદ રાખો)

Input Amplifier/Attenuator → Schmitt Trigger → Main Gate → Counter (DCU) → Display (+ Time Base Generator = Crystal Oscillator, Gate ને Control કરે)

- f = N (Pulse Count)/T (Gate Time)
- High Frequency → **Frequency Mode**
- Low Frequency → **Period Mode** (વધુ Accurate)
- **±1 Count Error**: Gate ના Random Open/Close ના કારણે; **ઘટાડવા Gate Time વધારો**

📘 ભાગ M: TRANSDUCERS

Active vs Passive

Point	Active	Passive
External Power	જરૂરી નથી	જરૂરી (Excitation)
ઉદાહરણ	Thermocouple, Piezoelectric, Photovoltaic	Strain Gauge, LVDT, Capacitive, RTD

Master Table

Transducer	Physical Quantity	Output/Principle
Strain Gauge	Force/Pressure	$\Delta R/R = G \times \varepsilon$ (Resistance)
RTD	Temperature	Resistance વધે (+ve Temp Coeff)
Thermistor	Temperature	Resistance ઘટે (−ve Temp Coeff, Non-linear)
Potentiometer	Displacement	Resistance/Voltage
LVDT	Linear Displacement	Differential Voltage (Null=0 at center)
Capacitive	Pressure/Level/Humidity	Capacitance (d, A કે ε બદલાય)
Piezoelectric	Vibration/Force (Dynamic only)	Voltage — $Q = d \times F$ (Active)
Thermocouple	Temperature (High)	Seebeck Effect, Voltage (Active)
Photovoltaic/LDR	Light	Voltage/Resistance
Hall Effect	Magnetic Field/Current	$VH = K \cdot I \cdot B/t$

🧠 Must-Remember

1. **RTD = +ve Coeff, Thermistor = −ve Coeff** (સૌથી common MCQ Confusion)
2. LVDT = Null Position પર Output = 0
3. Piezoelectric = Static Quantity Measure ના કરી શકે (Charge Leak થાય)
4. Thermocouple & Piezoelectric & Photovoltaic = **Active**; બાકી બધા લગભગ Passive

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📘 ભાગ N: COMPONENT TESTERS & ANALYZERS

Instrument	Measure/Test	Principle
Analog Multimeter	V, I, R	PMMC Galvanometer (Ohm scale Reverse!)
DMM	V,I,R,C,Diode,Continuity	ADC + Digital Display (High Input Impedance)
Transistor Tester	Type, Terminals, hFE	Diode Junction Test
LCR Meter	L, C, R, Q, D, ESR	AC signal + Phase measurement
ESR Meter	Capacitor Internal Resistance	SMPS Repair માટે Critical

Instrument	Measure/Test	Principle
Curve Tracer	V-I Characteristics	Varying Voltage vs Current Plot
SCR/Triac Tester	Latching, Triggering	Gate Trigger Test
Spectrum Analyzer	Frequency Components	Frequency Domain Display

DMM vs Analog Multimeter

Point	DMM	Analog
Accuracy	High (0.1-0.5%)	Low (2-5%)
Input Impedance	High (10MΩ+)	Low (Loading Effect)
Reading	Direct Digital	Pointer, Parallax possible

Diode/Capacitor Testing (Multimeter)

- Diode Forward: 0.5-0.7V (Si); Reverse: OL આવવું જોઈએ
- Capacitor Good: Needle $0 \rightarrow \infty$ (Charging Effect); S◌ીધું 0=Short; No Deflection=Open

CRO vs Spectrum Analyzer

Point	CRO	Spectrum Analyzer
X-axis	Time	Frequency
બતાવે	Waveform	Harmonics/Components

🎯 MASTER QUICK-REVISION (એક નજરમાં આખું Chapter)

Category	Key formula/fact
Superposition	Sources એક-એક active, Power માટે નહીં
Thevenin/Norton	$V_{th} = I_N \times R_N$, $R_{th} = R_N$
Max Power	$P_{max} = V_{th}^2 / 4R_{th}$
Tellegen	$\Sigma VI = 0$, Most General

Category	Key formula/fact
Resonance f0	1/(2π√LC)
Q-factor	ω0L/R
Filter fc	1/(π√LC)
Z0 (lossless)	√(L/C)
VSWR	(1+
Independent Loops	b−n+1
Schering	Capacitance+tan δ
Wien	Frequency
CRO Time base	Sawtooth
Freq. Counter	f=N/T
RTD	+ve coeff
Thermistor	−ve coeff
LVDT	Null=0

📰 CHAPTER-WISE CURRENT AFFAIRS (2026) — Exam Relevant

1. **National Frequency Allocation Plan (NFAP-2025)**: DoT દ્વારા 8.3kHz થી 3000GHz સુધીનું નવું Spectrum Management Plan જાહેર થયું; 6425-7125 MHz Band ને 5G/5G-Advanced/6G માટે Identify કરાયું છે.
2. **Spectrum Refarming**: સરકારે Defence, ISRO અને Information & Broadcasting મંત્રાલય પાસેથી 687 MHz Spectrum Refarm કરવાની મંજૂરી આપી, જેમાંથી 328 MHz તરત Release કરાયું — 5G Improve અને 6G તૈયારી માટે.
3. **6 GHz Band Debate**: Telecom Industry (COAI) સતત 6GHz Band ને IMT/Commercial Mobile Services માટે Delicense/Allocate કરવાની માંગ કરી રહી છે, જેથી 5G Data Capacity વધે.
4. **Bharat 6G Alliance & BharatNet**: ભારત 6G Research અને Fiber Connectivity Expansion પર કામ કરી રહ્યું છે; Mobile World Congress 2026 માં Communications Minister એ AI-driven Telecom Vision રજૂ કરી.
5. **5G Growth Data**: અંદાજે 2026ના અંત સુધીમાં આશરે 32% Mobile Subscription 5G પર હશે (Ericsson Estimate); 2031 સુધીમાં 1 Billion 5G Users નો Target.
6. **Satellite Broadband**: Starlink, Jio-SES, Eutelsat OneWeb જેવી Companies India માં Satellite-based Broadband Launch ની તૈયારીમાં છે, પણ Commercial Spectrum ના

Provisional Allocation ના કારણે Delay છે.

7. **Gujarat Police Wireless Recruitment 2026**: GPRB દ્વારા PSI (Wireless), Technical Operator અને PSI (MT) માટે 950+ જગ્યાઓ માટે ભરતી; Written Exam Electronics/Telecommunication Technical Knowledge પર આધારિત.

> 📌 નોંધ: Current Affairs Section સમય સાથે બદલાય છે — Exam પહેલાં તાજા સમાચાર (DoT/PIB વેબસાઈટ) જરૂર Check કરવા.

તૈયાર કરેલ: Wireless PSI Class-2/3 Exam Preparation માટે — Networks + Instruments & Measurements Chapter Summary